

T. Miguel Pegues

Evanston, IL | toumigpeg@gmail.com | 214-918-9478 | tmpegues.github.io | linkedin.com/in/tmpegues

Education

Northwestern University, Evanston, IL
M.S. in Robotics

Sept 2025 – Sept 2026 (expected)

California Institute of Technology, Pasadena, CA
B.S. in Mechanical Engineering

2018 – 2022

Skills

- **Robotics:** Computer vision, manipulation, motion planning, ROS, ROS 2, MoveIt 2, SLAM
- **Software Development:** C++, Python, C, Git, Linux, unit testing, embedded programming
- **Mechanical:** CAD (Creo, Onshape, SolidWorks), manual machining, design for assembly and manufacturing, 3D printing
- **Software Libraries:** CoppeliaSim, Gazebo, OpenCV, PyTorch, sklearn

Projects

Wearable Flexible Extra Finger

January - March 2026

- Created an tentacle-like extra finger that allows the user to manipulate a wider range of objects than they can naturally
- Designed human interfacing parts for sensors, motors, control boards, and the digit itself in Onshape and fabricated all parts by sewing, 3D printing, and laser cutting
- The range of motion of the digit, actuated by three motors, far exceeds that of a natural human finger and hand
- The digit is controlled by a hand-made glove with integrated flex sensors that interfaces with ROS 2 through a Pico 2 sending analog values to a Python ROS 2 node over USB or is read directly by an ADC connected to a Raspberry Pi 4
- Flex sensor values map to motor velocities, spooling and unspooling cables to flex the digit in all three directions

Dynamic Impedance Prosthetic Ankle

May 2026 - Present

- Designed ankle prosthesis with adjustable impedance to allow users to vary their gait speed at will
- Built and tested novel mechanisms (radial cantilevers, magnetic return, among others) for adjusting stiffness of torsion-spring-like rotational elements under high-load cyclic use
- Implemented gait detection algorithms using IMU and force sensing to modulate quasi-static mechanism to automatically adjust prosthesis properties to the user's speed

Bug Catching Robot

December 2025

- Wrote ROS 2 packages for monocular object detection and motion planning to use a Franka Emika Robot arm to catch and sort colored HexBugs based on visual user input with a team of four
- Focused on motion planning and execution for dynamic tracking of a moving target using MoveIt 2 and `ros2_control`, maximizing responsiveness by sending joint trajectories directly to the `joint_trajectory_controller`
- Designed and 3D printed parts for physical arena and supported camera calibration with Python, OpenCV, and ROS 2

Jenga Playing Robot

Winter - Spring 2022

- Designed and built a robot arm capable of playing Jenga with a team of four
- Machined metal robot parts and laser cut Jenga block markers and alignment tools after designing in SolidWorks
- My role focused writing and tuning the vision system with OpenCV to locate all 54 Jenga blocks in 3D space with millimeter precision, then sending coordinates in ROS message to the motion planning node

Work Experience

Mechanical Structures Engineer

July 2022 - July 2025

Whirlpool Corporation – Benton Harbor, MI

- Drove Cost & Quality and Product Development projects, designing plastic and metal components for all refrigeration architectures to improve cost and customer satisfaction
- Coordinated with domestic and international suppliers to ensure on-time delivery of new and refurbished parts
- Managed verification and validation testing for individual components, systems, and products
- Provided on-site support to international colleagues for lab, model shop, marketing, and manufacturing evaluations